

Supplementary Material: Detailed per-split results for “Beyond Random Splits: A Leakage-Resistant Evaluation Protocol for Cross-Source Deceptive Email Detection”

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TABLE S1: MAIN PERFORMANCE BY MODEL AND SPLIT DESIGN

This table reproduces in full the per-model, per-protocol, per-holdout detail that supports Table IV in the main manuscript. SD = source-disjoint holdout (six candidates: trec5, trec6, trec7, trec5+trec6, trec5+trec7, trec6+trec7). RCD-p = sources-pooled cluster-disjoint control matched to each SD holdout on training size, test size, per-class counts, and random seed (no source held out). Rand = stratified random split (three seeds; one row shown per model, median over the six matched holdouts). Sec = independent secondary corpus (ealvaradob phishing subset, 20,069 messages). CI is a stratified-bootstrap 95% interval over 1,000 replicates. Values are read from the canonical metrics CSV `outputs/runs/20260811T162016Z_d2881b85/metrics/all_metrics.csv`; the run ID is in the header comment.

REPRODUCIBILITY NOTE

The full per-split values in Table S1 are generated from the canonical metrics file `outputs/runs/20260811T162016Z_d2881b85/metrics/all_metrics.csv` and the result macros in `paper/generated/result_macros.tex`. The supplementary file does not introduce any new values; it re-displays the same numbers in full per-split form for reviewers who require them.

TABLE I
FULL PER-SPLIT PERFORMANCE FOR ALL SIX MODELS ACROSS ALL FOUR PROTOCOLS. NUMERIC VALUES ARE GENERATED FROM
GENERATED/RESULT_MACROS.TEX (SEE MAIN MANUSCRIPT FOR FULL MACRO DEFINITIONS).

Model	Design	Split	Macro-F1 (95% CI)	Prec.+	Rec.+	MCC
M2	SD	holdout_trec5	0.868 [0.865, 0.871]	0.773	0.967	0.756
A2	SD	holdout_trec5	0.854 [0.851, 0.857]	0.751	0.968	0.732
M3	SD	holdout_trec5	0.764 [0.760, 0.768]	0.882	0.563	0.571
M1	SD	holdout_trec5	0.841 [0.838, 0.844]	0.734	0.964	0.710
A1	SD	holdout_trec5	0.825 [0.822, 0.828]	0.709	0.976	0.691
M4	SD	holdout_trec5	0.772 [0.769, 0.775]	0.655	0.945	0.597
M2	SD	holdout_trec5_trec6	0.685 [0.682, 0.689]	0.542	0.982	0.508
A2	SD	holdout_trec5_trec6	0.671 [0.667, 0.675]	0.531	0.984	0.492
M3	SD	holdout_trec5_trec6	0.416 [0.412, 0.419]	0.348	0.649	-0.072
M1	SD	holdout_trec5_trec6	0.665 [0.662, 0.669]	0.527	0.988	0.488
A1	SD	holdout_trec5_trec6	0.649 [0.646, 0.652]	0.516	0.991	0.472
M4	SD	holdout_trec5_trec6	0.567 [0.563, 0.570]	0.466	0.979	0.370
M2	SD	holdout_trec5_trec7	0.894 [0.892, 0.896]	0.875	0.908	0.788
A2	SD	holdout_trec5_trec7	0.886 [0.884, 0.889]	0.860	0.912	0.774
M3	SD	holdout_trec5_trec7	0.775 [0.772, 0.777]	0.772	0.755	0.550
M1	SD	holdout_trec5_trec7	0.882 [0.880, 0.884]	0.861	0.899	0.764
A1	SD	holdout_trec5_trec7	0.871 [0.869, 0.874]	0.841	0.903	0.745
M4	SD	holdout_trec5_trec7	0.838 [0.836, 0.841]	0.878	0.774	0.681
M2	SD	holdout_trec6	0.895 [0.891, 0.901]	0.759	0.960	0.802
A2	SD	holdout_trec6	0.900 [0.895, 0.905]	0.769	0.958	0.810
M3	SD	holdout_trec6	0.751 [0.744, 0.760]	0.679	0.555	0.508
M1	SD	holdout_trec6	0.917 [0.912, 0.922]	0.809	0.959	0.840
A1	SD	holdout_trec6	0.926 [0.922, 0.930]	0.834	0.952	0.855
M4	SD	holdout_trec6	0.856 [0.851, 0.862]	0.690	0.931	0.729
M2	SD	holdout_trec6_trec7	0.844 [0.841, 0.847]	0.793	0.907	0.695
A2	SD	holdout_trec6_trec7	0.852 [0.849, 0.855]	0.809	0.902	0.709
M3	SD	holdout_trec6_trec7	0.621 [0.617, 0.625]	0.578	0.792	0.282
M1	SD	holdout_trec6_trec7	0.854 [0.851, 0.856]	0.811	0.901	0.712
A1	SD	holdout_trec6_trec7	0.869 [0.867, 0.872]	0.851	0.879	0.739
M4	SD	holdout_trec6_trec7	0.774 [0.771, 0.777]	0.725	0.845	0.558
M2	SD	holdout_trec7	0.898 [0.896, 0.901]	0.918	0.897	0.796
A2	SD	holdout_trec7	0.903 [0.900, 0.905]	0.930	0.893	0.806
M3	SD	holdout_trec7	0.592 [0.588, 0.597]	0.624	0.775	0.209
M1	SD	holdout_trec7	0.886 [0.884, 0.890]	0.905	0.891	0.773
A1	SD	holdout_trec7	0.886 [0.883, 0.889]	0.907	0.887	0.772
M4	SD	holdout_trec7	0.842 [0.839, 0.846]	0.890	0.816	0.688
M2	Rand	random seed42	0.986 [0.984, 0.987]	0.982	0.987	0.972
A2	Rand	random seed42	0.984 [0.982, 0.985]	0.979	0.985	0.968
M3	Rand	random seed42	0.804 [0.798, 0.810]	0.864	0.686	0.622
M1	Rand	random seed42	0.982 [0.980, 0.984]	0.976	0.985	0.964
A1	Rand	random seed42	0.981 [0.979, 0.983]	0.974	0.984	0.962
M4	Rand	random seed42	0.959 [0.956, 0.961]	0.955	0.953	0.917
M2	Sec	secondary test	0.858 [0.854, 0.862]	0.759	0.931	0.729
A2	Sec	secondary test	0.887 [0.884, 0.891]	0.836	0.893	0.776
M3	Sec	secondary test	0.281 [0.280, 0.283]	0.381	0.991	-0.016
M1	Sec	secondary test	0.881 [0.877, 0.885]	0.791	0.946	0.772
A1	Sec	secondary test	0.896 [0.892, 0.900]	0.839	0.913	0.794
M4	Sec	secondary test	0.787 [0.783, 0.793]	0.665	0.903	0.605
M2	RCD-p	pooled holdout trec5	0.983 [0.982, 0.984]	0.976	0.984	0.966
A2	RCD-p	pooled holdout trec5	0.982 [0.981, 0.984]	0.974	0.982	0.964
M3	RCD-p	pooled holdout trec5	0.797 [0.794, 0.801]	0.843	0.648	0.605
M1	RCD-p	pooled holdout trec5	0.978 [0.977, 0.979]	0.970	0.977	0.957
A1	RCD-p	pooled holdout trec5	0.977 [0.975, 0.978]	0.967	0.976	0.954
M4	RCD-p	pooled holdout trec5	0.956 [0.955, 0.958]	0.940	0.954	0.913
M2	RCD-p	pooled holdout trec6	0.980 [0.977, 0.983]	0.952	0.985	0.961
A2	RCD-p	pooled holdout trec6	0.978 [0.975, 0.981]	0.947	0.984	0.957
M3	RCD-p	pooled holdout trec6	0.801 [0.793, 0.809]	0.702	0.669	0.620
M1	RCD-p	pooled holdout trec6	0.973 [0.970, 0.977]	0.936	0.982	0.946
A1	RCD-p	pooled holdout trec6	0.972 [0.969, 0.975]	0.934	0.980	0.944
M4	RCD-p	pooled holdout trec6	0.944 [0.940, 0.948]	0.872	0.959	0.893
M2	RCD-p	pooled holdout trec7	0.985 [0.984, 0.986]	0.989	0.984	0.970
A2	RCD-p	pooled holdout trec7	0.984 [0.983, 0.985]	0.989	0.983	0.968
M3	RCD-p	pooled holdout trec7	0.785 [0.781, 0.788]	0.911	0.685	0.600
M1	RCD-p	pooled holdout trec7	0.981 [0.980, 0.982]	0.985	0.981	0.961
A1	RCD-p	pooled holdout trec7	0.979 [0.978, 0.981]	0.984	0.980	0.958
M4	RCD-p	pooled holdout trec7	0.953 [0.951, 0.955]	0.978	0.939	0.907